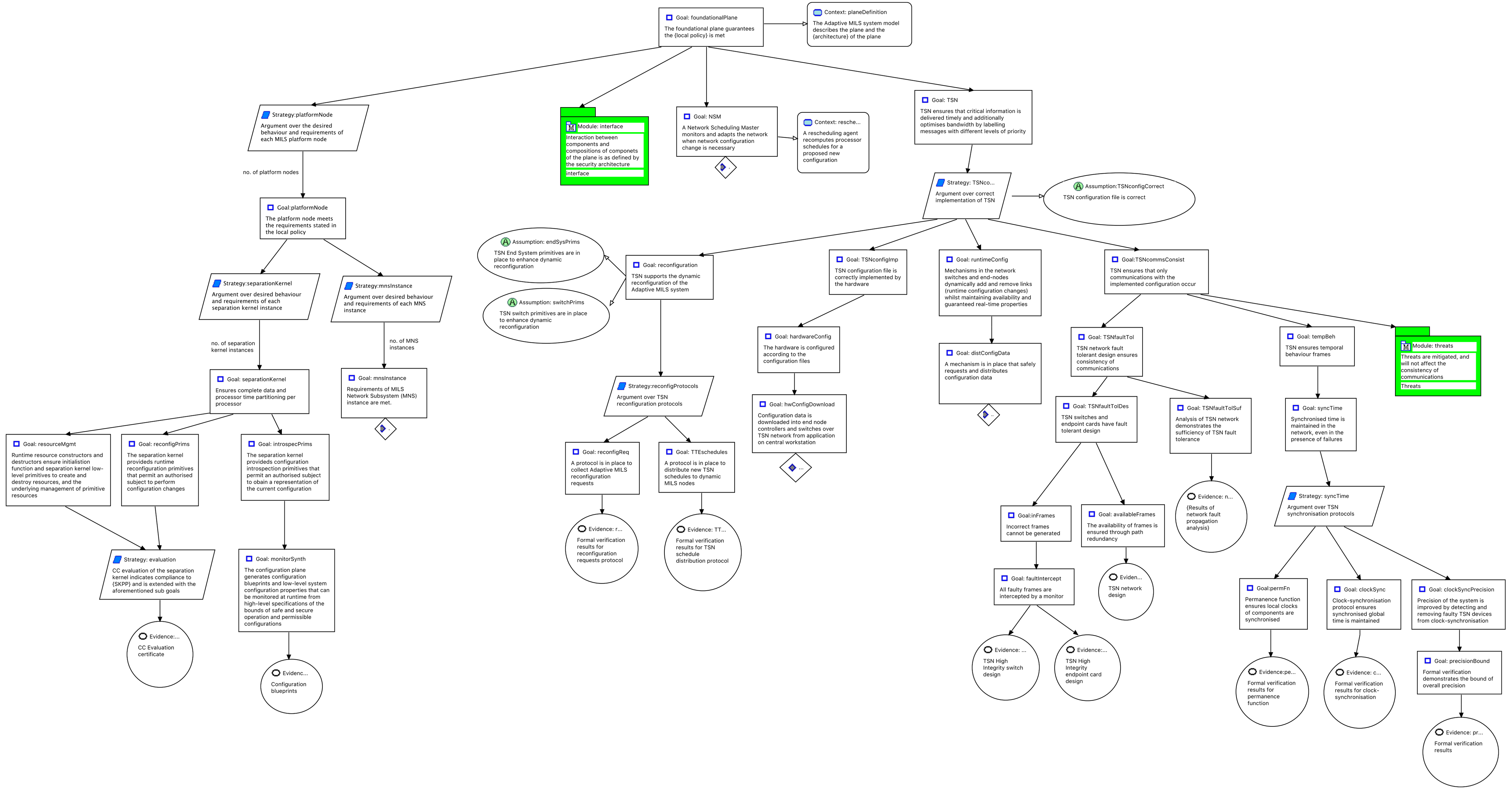
Goal: foundationalPlane
The foundational plane guarantees the {local policy} is met
Context: planeDefinition
The Adaptive MILS system model describes the plane and the {architecture} of the plane
Strategy:platformNode
Argument over the desired behaviour and requirements of each MILS platform node
no. of platform nodes
Goal:platformNode
The platform node meets the requirements stated in the local policy
Strategy:separationKernel
Argument over desired behaviour and requirements of each separation kernel instance
Strategy:mnsInstance
Argument over desired behaviour and requirements of each MNS instance
no. of separation kernel instances
no. of MNS instances
Goal: separationKernel
Ensures complete data and processor time partitioning per processor
Goal: mnsInstance
Requirements of MILS Network Subsystem (MNS) instance are met.
Goal: resourceMgmt
Runtime resource constructors and destructors ensure initialistion function and separation kernel low-level primitives to create and destroy resources, and the underlying management of primitive resources
Goal: reconfigPrims
The separation kernel provideds runtime reconfiguration primitives that permit an authorised subject to perform configuration changes
Goal: introspecPrims
The separation kernel provideds configuration introspection primitives that permit an authorised subject to obain a representation of the current configuration
Strategy: evaluation
CC evaluation of the separation kernel indicates compliance to {SKPP} and is extended with the aforementioned sub goals
Evidence:...
CC Evaluation certificate
Goal: monitorSynth
The configuration plane generates configuration blueprints and low-level system configuration properties that can be monitored at runtime from high-level specifications of the bounds of safe and secure operation and permissible configurations
Evidenc...
Configuration blueprints
Module: interface
Interaction between components and compositions of componets of the plane is as defined by the security architecture
interface
Goal: NSM
A Network Scheduling Master monitors and adapts the network when network configuration change is necessary
Context: resche...
A rescheduling agent recomputes processor schedules for a proposed new configuration
Goal: TSN
TSN ensures that critical information is delivered timely and additionally optimises bandwidth by labelling messages with different levels of priority
Strategy: TSNco...
Argument over correct implementation of TSN
Assumption:TSNconfigCorrect
TSN configuration file is correct
Assumption: endSysPrims
TSN End System primitives are in place to enhance dynamic reconfiguration
Assumption: switchPrims
TSN switch primitives are in place to enhance dynamic reconfiguration
Goal: reconfiguration
TSN supports the dynamic reconfiguration of the Adaptive MILS system
Strategy:reconfigProtocols
Argument over TSN reconfiguration protocols
Goal: reconfigReq
A protocol is in place to collect Adaptive MILS reconfiguration requests
Evidence: r...
Formal verification results for reconfiguration requests protocol
Goal: TTEschedules
A protocol is in place to distribute new TSN schedules to dynamic MILS nodes
Evidence: TT...
Formal verification results for TSN schedule distribution protocol
Goal: TSNconfigImp
TSN configuration file is correctly implemented by the hardware
Goal: hardwareConfig
The hardware is configured according to the configuration files
Goal: hwConfigDownload
Configuration data is downloaded into end node controllers and switches over TSN network from application on central workstation
Goal: runtimeConfig
Mechanisms in the network switches and end-nodes dynamically add and remove links {runtime configuration changes} whilst maintaining availability and guaranteed real-time properties
Goal: distConfigData
A mechanism is in place that safely requests and distributes configuration data
Goal:TSNcommsConsist
TSN ensures that only communications with the implemented configuration occur
Goal: TSNfaultTol
TSN network fault tolerant design ensures consistency of communications
Goal: tempBeh
TSN ensures temporal behaviour frames
Module: threats
Threats are mitigated, and will not affect the consistency of communications
Threats
Goal: TSNfaultTolDes
TSN switches and endpoint cards have fault tolerant design
Goal: TSNfaultTolSuf
Analysis of TSN network demonstrates the sufficiency of TSN fault tolerance
Goal: syncTime
Synchronised time is maintained in the network, even in the presence of failures
Evidence: n...
{Results of network fault propagation analysis}
Goal:inFrames
Incorrect frames cannot be generated
Goal: availableFrames
The availability of frames is ensured through path redundancy
Eviden...
TSN network design
Strategy: syncTime
Argument over TSN synchronisation protocols
Goal: faultIntercept
All faulty frames are intercepted by a monitor
Evidence: ...
TSN High Integrity switch design
Evidence:...
TSN High Integrity endpoint card design
Goal:permFn
Permanence function ensures local clocks of components are synchronised
Goal: clockSync
Clock-synchronisation protocol ensures synchronised global time is maintained
Goal: clockSyncPrecision
Precision of the system is improved by detecting and removing faulty TSN devices from clock-synchronisation
Evidence:pe...
Formal verification results for permanence function
Evidence: c...
Formal verification results for clock-synchronisation
Goal: precisionBound
Formal verification demonstrates the bound of overall precision
Evidence: pr...
Formal verification results